Thesis checklist

Content, clarity, and motivation issues only. Citation checking is deferred. Source line numbers may shift.

3.2.2 Bregman and beta-divergences

- □ **Repair the beta-divergence domain.** Reconcile the real power t^(1+beta) and regularity assumptions for arbitrary real beta > 0, including the separate quadratic case. [chapter 175-183]
- □ **Justify the symmetry characterization.** Establish that the stated assumptions justify the functional symmetry characterization, including the Legendre-type condition, or give a separate proof. [chapter 159-163]
- □ **Correct the scalar test functions.** Use the scalar t after defining p_t in the affine-invariance argument and justify the scalar conclusion in the symmetry proof. [chapter 170-172]

3.5.2 Proposition 9: proof

- □ **Introduce notation before it is used.** Define transformed coordinates and endpoints, m(t), q_0, sigma_N, and the maximizing-index sequence. Keep N and subsequence indices consistent. [chapter 620-649]
- □ **Justify the boundary row-sum bound.** The upper-bound argument permits an endpoint limit but refers to Proposition 8, whose statement assumes an interior limit. Explain the boundary case separately. [chapter 627-645; 575-580]
- □ **Correct the Rayleigh-bound notation.** Use b_N consistently and identify the correct index set. In the pair-count explanation, j is outside I_N; up to k indices can occur on each side. [chapter 666-675]
- □ **Correct the sign in the E_N bound.** Replace u-v by the interval length v-u in the lower bound for sigma_N times the number of indices, and check the limit. [chapter 677]
- □ **Connect the final index to the stated maximizer.** Identify the final i_N with the maximizing index used at the start and with k_N in the proposition, and make the limit argument explicit. [chapter 616-621, 681]
- □ **Resolve the empty appendix heading.** The proof now appears in the main chapter, leaving its appendix subsection empty. Resolve that heading and any relevant references. [chapter 620-682; appendix 495]

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3.5.3 Energy functional and bandwidth selection

[ ] **Qualify the connection to classical bandwidth selection.** Separate convergence of the spectral selector to a sample energy minimizer, comparison of leading-term criteria, and any claim about minimizers of the exact expected energy or MISE. State the assumptions and bandwidth domain for each. [chapter 691-695, 801-817]

[ ] **Check the Gaussian formula and endpoint cases.** Label the explicit pair-sum formula and define the integration bounds in the normal-moment identity. The small-bandwidth coefficient shown assumes distinct observations; allow for ties or state the assumption. Identify the range of r used for the limits. [chapter 697-717]

[ ] **Finish the compact-interval minimizer argument.** Replace the empty equation and theorem references; explain positivity and nonconstancy of the analytic energy before using the identity theorem. Distinguish the finite set of constrained minimizers from a global minimizer on all positive bandwidths. [chapter 717-719]

[ ] **Complete the roughness-ratio derivation.** Justify the heat-equation derivative identity, positivity of the denominator, and both limits of B. State the range of r and make the sign-change argument for local minima and maxima precise, including degenerate stationary points. [chapter 721-727]

[ ] **Tighten the high-probability proof.** State the kernel and density regularity assumptions that justify the convolution limits, and repair the absolute-value expression in the probability bound. Clarify that the claim concerns existence of a local minimum, not its uniqueness or its location on a fixed interval. [chapter 729-741]

[ ] **Make the numerical minimization claim defensible.** A single detected local minimum and the endpoints need not certify a global minimum: explain how all competing stationary points are found or qualify the result as numerical. [chapter 743-745]

3.5.3 Energy functional and bandwidth selection

[ ] **Explain the benchmark and complete its table.** Give the bandwidth interval, geometric grid, timing scope, reference used for relative error, and meaning of SE; add a descriptive caption. The reference is a checked numerical minimum, not an exhaustive certificate. [chapter 748-775]

[ ] **Restrict the runtime and convergence conclusions.** Describe the timing advantage as specific to the measured implementation and sample. State both lambda_N going to infinity and lambda_N / N^2 going to zero, and avoid claiming that the result guarantees monotone finite-N errors or makes spectral methods universally redundant. [chapter 750, 777-781]

[ ] **Correct the exact expected derivative norm.** Re-derive the diagonal and off-diagonal terms: the diagonal squared norm scales as h^-3, and the exact variance includes subtraction of the squared mean term. Then reconcile the exact expression with the leading-term approximation. [chapter 785-789, 801-804]

[ ] **Repair and justify the small-bandwidth limit.** Specify suitable kernel and density assumptions or state the Gaussian case; replace the invalid squared-norm inequality by a valid argument using continuity of the norm. Track the kernel second-moment factor consistently in the leading term. [chapter 791-803]

[ ] **Recalculate the AMQ-to-AMISE ratio.** Place the entire ratio of constants under the fifth root and check the Gaussian specialization and r = 13/14. State that this compares minimizers of leading-term approximations; prove any stronger equivalence for exact minimizers separately. [chapter 806-817]

[ ] **Polish language and cross-references.** Correct typos and capitalization, define symbols before use, improve the section title and transitions, and check figure references and the wording of the spectral penalty. Remove the two polishing markers after the work is complete. [chapter 691-817]

Review basis: bandwidth-selection chapter, lines 691-817; supporting project calculations in output/research/gaussian_qhat_properties_2026-10-29/report.md; benchmark report in Code/Code_kde/outputs/brent_log_spectral_m100_comparison/report.md. Mathematical corrections are review findings to verify when revising the thesis.